NETFLIX PRIZE · PERSONALIZED CONTENT DISCOVERY

Recommendation Systems for Personalized Content Discovery

Bias · Matrix Factorization · Item-based Collaborative Filtering

Mandatory metrics: RMSE & MAP@10 | relevance threshold = 3.5★

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The Problem & The Data

Goal: learn preferences from past ratings and surface the right Top-10 unseen titles per user.

Two coupled tasks that don't always agree:

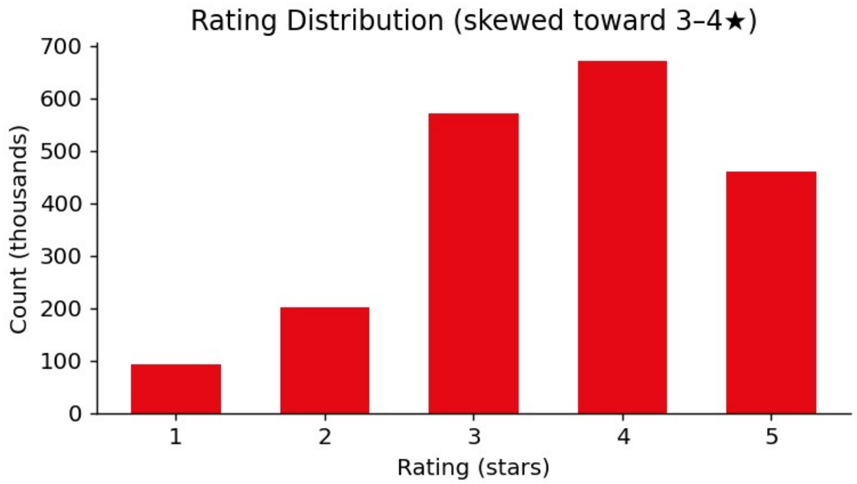
- **Rating prediction** → measured by RMSE
- **Ranking** → measured by MAP@10

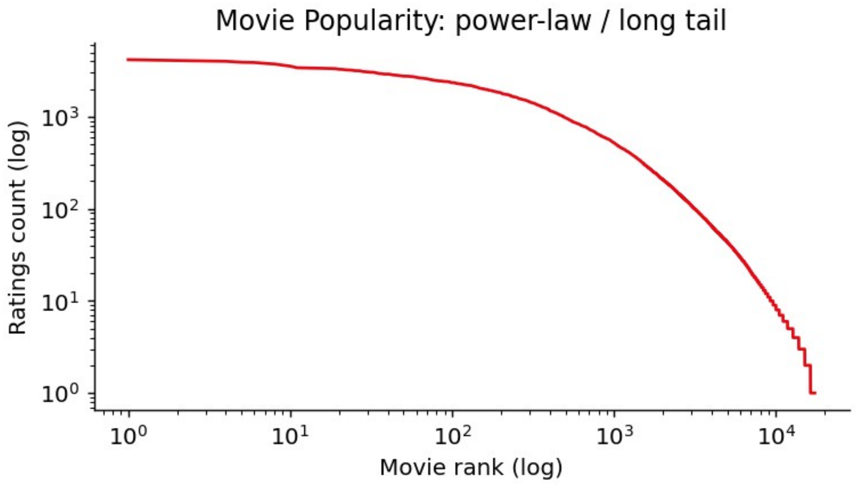
> Core challenges: extreme sparsity, long-tail popularity, positivity bias, and cold-start.

Working set (k-core)	
Ratings	629,679
Users	28,411
Movies	6,823
Density	0.325%
Mean rating	3.51★

Sampled from the full 100M-rating corpus via reservoir sampling, then k-core filtered (users ≥ 15, movies ≥ 10).

What the Data Told Us





- **Positivity bias** — ~85% of ratings are 3–5★; a model must beat the "predict the mean" trap.
- **Power-law popularity** — a few blockbusters dominate, making popularity a deceptively strong baseline.
- **Temporal growth** — rating volume rises sharply, so we use a temporal (past→future) split, not random.

Three Models, One Comparison

Model	Prediction	Strength
Baseline (Bias)	μ + b_u + b_i	Strong, cheap RMSE floor; popularity ranker
Matrix Factorization	μ + b_u + b_i + p_u·q_i	Latent-factor personalization (SGD)
Item-based CF	cosine item neighbours	Explainable, diversity-oriented

Evaluation protocol. Temporal 80/20 per-user split · relevance = rating ≥ 3.5 · Top-10 by scoring all items and masking seen ones · reported on both all-items and 100-sampled-negative protocols.

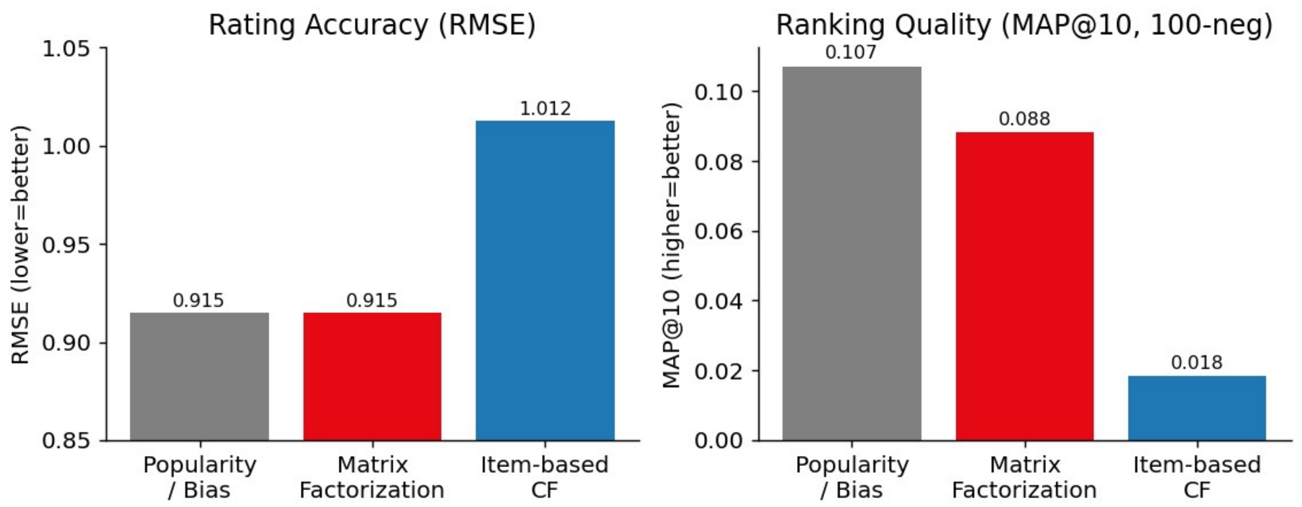
Results

Rating accuracy

Model	RMSE
Bias	**0.915**
MF	0.915
Item-CF	1.012

Ranking (100-neg)

Model	MAP@10	HR@10
Popularity	**0.107**	**0.50**
MF	0.088	0.45
Item-CF	0.018	0.13

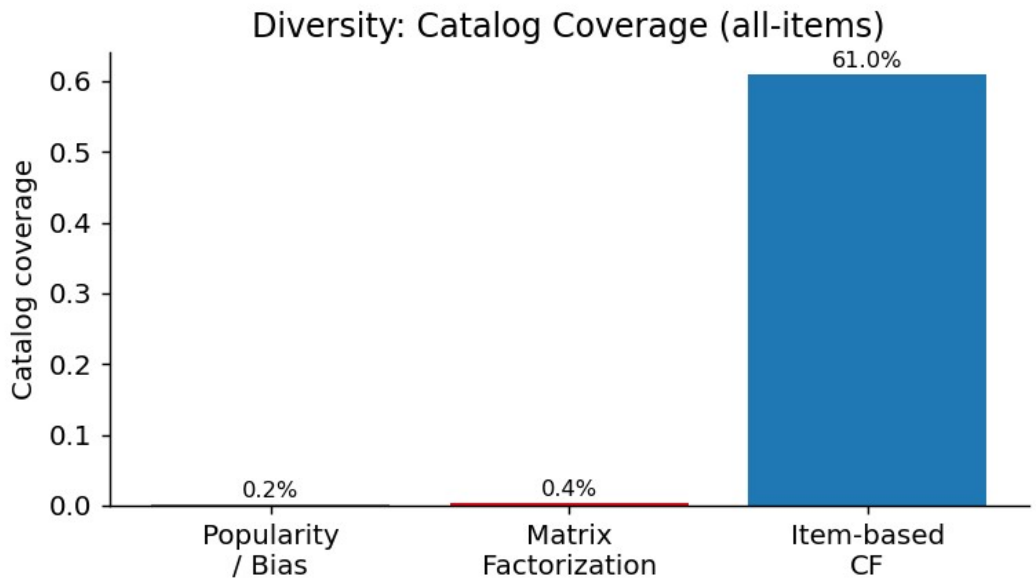


The Central Trade-off

Models that tie on RMSE behave completely differently when ranking.

- **Popularity** wins hit-rate — but covers <1% of the catalog.
- **MF** personalizes at near-equal quality.
- **Item-CF** reaches 61% coverage — far more diverse — at an accuracy cost.

No single model wins every metric. The right choice depends on the objective: hit-rate, personalization, or discovery.



Recommendations in Action

MF Top-10 for a user who loved epics & acclaimed series — it recovered a held-out relevant title (✓):

#	Recommended
1	The Simpsons: Season 5
2	LOTR: Return of the King (Ext.)
3	LOTR: The Two Towers (Ext.)
5	**Family Guy Collection ✓**
8	The Shawshank Redemption

Explainability. Item-CF supports natural "because you watched X" rationales — though sparse co-ratings make those similarities noisy, which is exactly why its ranking lags.

Key Insights & What's Next

INSIGHTS

- RMSE ≠ recommendation quality.
- Popularity bias is strong but kills diversity.
- MF is the best balanced single choice.
- Sparsity punishes memory-based CF.

FUTURE WORK

- Scale to the full 100M ratings.
- Implicit & ranking-native models (ALS, BPR).
- Hybrid / content features for cold-start.
- Diversity-aware re-ranking; neural CF.

Reproducible notebook · fixed seed 42 · 2,500 evaluation users